# Takuya Koriyama

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## Research Interests

I'm broadly interested in statistics, ML, and probability theory. My previous research has focused on high-dimensional statistics, high-dimensional probability/geometry, and combinatorial stochastic processes. More recently, I'm working on applied AI research, including AI alignment and model evaluation.

## Education

**University of Chicago, Booth School of Business** 2024–present
PhD student, Econometrics and Statistics. Advisor: Tengyuan Liang

**University of Tokyo** 2018–2022
BE, Mathematical Engineering

## Selected Publications

### A Markov Chain Approach to Preference Alignment

**Takuya Koriyama** and Tengyuan Liang
Preprint, arXiv:2606.22652, 2026 [arXiv]

Honorable Mention for New Solution Concept, ICML 2026 NExT-Game Workshop

Introduces Markov Chain from Human Feedback to align generative models directly from pairwise preferences. Establishes convergence guarantees that adapt to non-transitive preferences and connects the method to reinforcement learning and game-theoretic alignment.

### Precise Asymptotics of Bagging Regularized M-estimators

**Takuya Koriyama**, Pratik Patil, Jin-Hong Du, Kai Tan, and Pierre C. Bellec
**The Annals of Statistics**, 2026 [arXiv] [Journal]

Selected for the ICML Journal-to-Conference track

Derives precise prediction-error formulas and a consistent risk estimator for ensembles trained on overlapping subsamples. Shows how jointly tuning subsample size, ensemble size, and regularization can improve predictive performance.

### Asymptotic Inference for Exchangeable Gibbs Partitions

**Takuya Koriyama**
**Stochastic Processes and their Applications**, 2026 [arXiv] [Journal]

Develops parameter estimation and predictive uncertainty quantification for random partition models, including asymptotically valid confidence intervals and error quantification for predicting allocation to existing or new groups.

### Denoising Diffusions with Optimal Transport: Localization, Curvature, and Multi-Scale Complexity

Tengyuan Liang, Kulunu Dharmakeerthi, and **Takuya Koriyama**
**Transactions on Machine Learning Research**, 2026 [arXiv] [Journal]

Selected for Journal-to-Conference certification

Characterizes score-based denoising through optimal transport and quantifies how data geometry across noise scales determines denoising difficulty. Identifies bottleneck noise levels for complex, non-log-concave distributions.

### Corrected generalized cross-validation for finite ensembles of penalized estimators

Pierre C. Bellec, Jin-Hong Du, **Takuya Koriyama**, Pratik Patil, and Kai Tan (alphabetical)
**Journal of the Royal Statistical Society: Series B**, 87(2), 289–318, 2025 [arXiv] [Journal]

Develops corrected generalized cross-validation for estimating prediction error in ensembles of penalized least-squares models. Provides statistical guarantees for an efficient evaluation and tuning method that avoids holding out training data or repeatedly refitting models.

## Additional Publications

### Asymptotics of resampling without replacement in robust and logistic regression
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
**Bernoulli**, 32(4), 3063–3087, 2026 [arXiv] [Journal]

### Asymptotic mixed normality of maximum likelihood estimator for Ewens–Pitman partition
**Takuya Koriyama**, Takeru Matsuda, and Fumiyasu Komaki
**Advances in Applied Probability**, 58(1), 214–234, 2026 [arXiv] [Journal]

### Phase transitions for the existence of unregularized M-estimators in single index models
**Takuya Koriyama** and Pierre C. Bellec
**International Conference on Machine Learning (ICML)**, 2025 [arXiv] [Proceedings]

### Error estimation and adaptive tuning for unregularized robust M-estimator
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
**Journal of Machine Learning Research**, 26(16), 1–40, 2025 [arXiv] [Journal]

### Fully Data-driven Normalized and Exponentiated Kernel Density Estimator with Hyvärinen Score
Shunsuke Imai, **Takuya Koriyama**, Shouto Yonekura, Shonosuke Sugasawa, and Yoshihiko Nishiyama
**Journal of Business & Economic Statistics**, 43(1), 110–121, 2025 [arXiv] [Journal]

### Existence of solutions to the nonlinear equations characterizing the precise error of M-estimators
Pierre C. Bellec and **Takuya Koriyama** (alphabetical)
Preprint, arXiv:2312.13254, 2023; submitted [arXiv]

## Experience

**Student Researcher, RIKEN Center for Brain Science** 2021–2022
Theory of combinatorial stochastic processes and its application to network data analysis

## Awards and Honors

- Honorable Mention for New Solution Concept, ICML NExT-Game Workshop, 2026.
- IMS Hannan Graduate Student Travel Award, 2025.
- Travel Award, Yale FDS Workshop Honoring Andrew Barron, 2024.
- Student Travel Award, IMS International Conference on Statistics and Data Science, 2022.
- Best Presentation Award, 16th Japan Statistical Society Spring Meeting, 2022.
- Japan Statistical Society Certificate Director's Award, 2022.

## Academic Service

**Reviewer:** ICML; The Annals of Statistics; Journal of the American Statistical Association; Information and Inference: A Journal of the IMA.